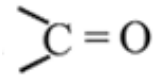
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Carboxylic acid group does not give the usual addition and elimination reactions of aldehydes and ketones because

(A) O-H bond is more polar than



group

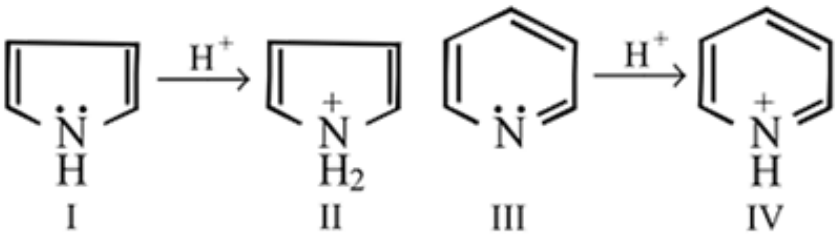
(B) carboxylate ion gets ionised

(C) carboxylate ion gets stabilised by resonance

(D) it exists as - COOH and there is no carbonyl group

Q.2

Pyrrole and pyridine both are basic and form salts with acids?



Which of the following statement is true regarding the aromatic character of the four species?

(A) All the four are aromatic

(B) I, III and IV are aromatic

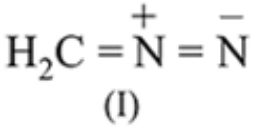
(C) I, II and III are aromatic

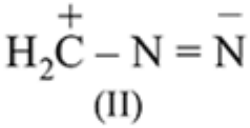
(D) I and III are aromatic

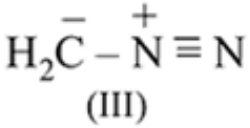
Q.3

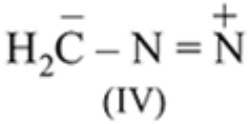
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The correct stability order of the following resonance structures is


(I)


(II)


(III)


(IV)

(A) (I) > (II) > (IV) > (III)

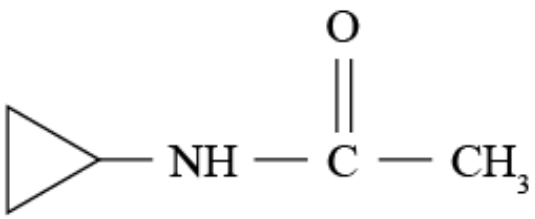
(B) (I) > (III) > (II) > (IV)

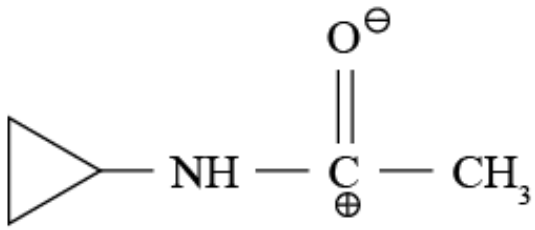
(C) (II) > (I) > (III) > (IV)

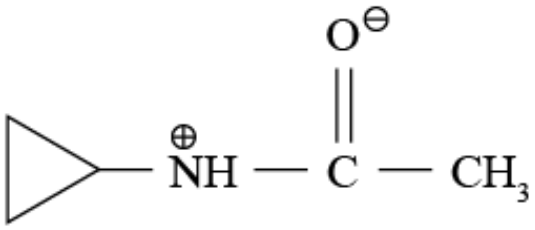
(D) (III) > (I) > (IV) > (II)

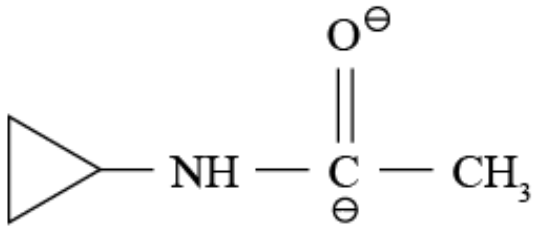
Q.4

Correct stability order of resonating structure is:

(P) 

(R) 

(Q) 

(S) 

(A) P > Q > R > S

(B) P > Q > S > R

(C) P > R > Q > S

(D) P > S > Q > R

Q.5

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Pick up the correct statement regarding the following resonating structures of the anilinium ion



I II

(A) Structure II is not acceptable because carbonium ions are less stable than ammonium ions

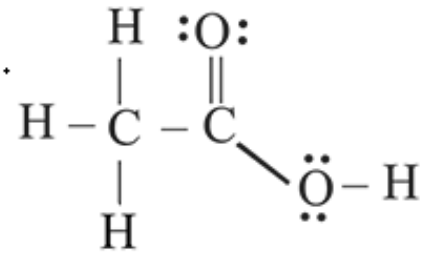
(B) II is not acceptable because it is nonaromatic

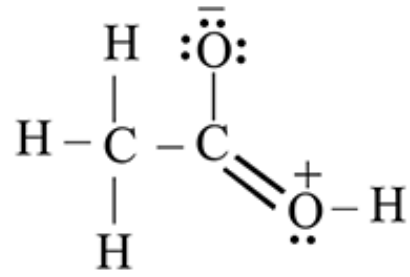
(C) II is not acceptable because here nitrogen has 10 valence electrons

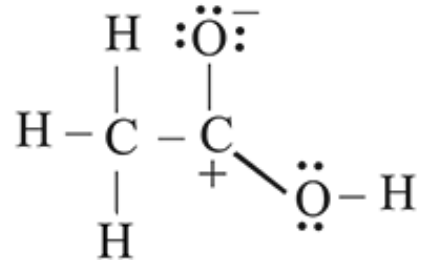
(D) II is an acceptable canonical structure.

Q.6

Which of the following resonance structure is lowest in energy?

(A) 

(B) 

(C) 

(A) A

(B) B

(C) C

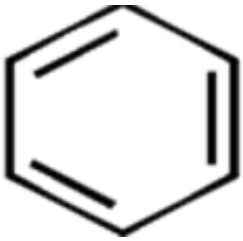
(D) All have same energy

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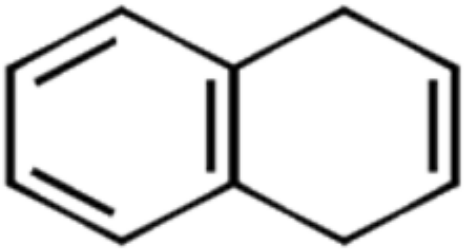
Q.7

All the hydrocarbons shown are very weak acids. One, however, is far more acidic than the others. Which one is the strongest acid?

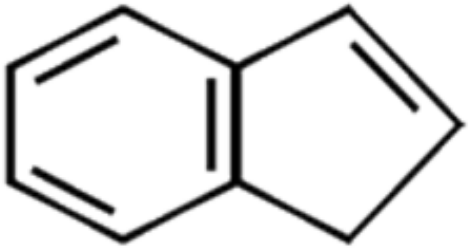
(A)



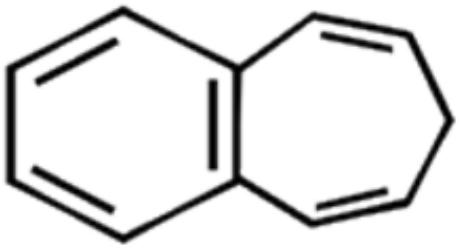
(B)



(C)



(D)



Q.8

An ortho-para directing group

(A) activates ortho and para positions and deactivates the meta position

(B) activates meta position and deactivates ortho and para positions

(C) activates ortho and para positions more than the meta position

(D) deactivates ortho and para positions less than the meta position

Q.9

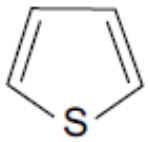
The nonaromatic compound along the following is

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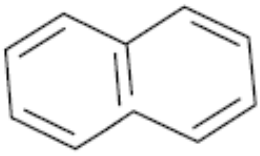
(A)



(B)



(C)



(D)



Q.10

The correct order of acid strength of the following compounds

A. Phenol

B. p-Cresol

C. m-Nitrophenol

D. p - Nitrophenol is

(A) $\mathrm{C} > \mathrm{B} > \mathrm{A} > \mathrm{D}$

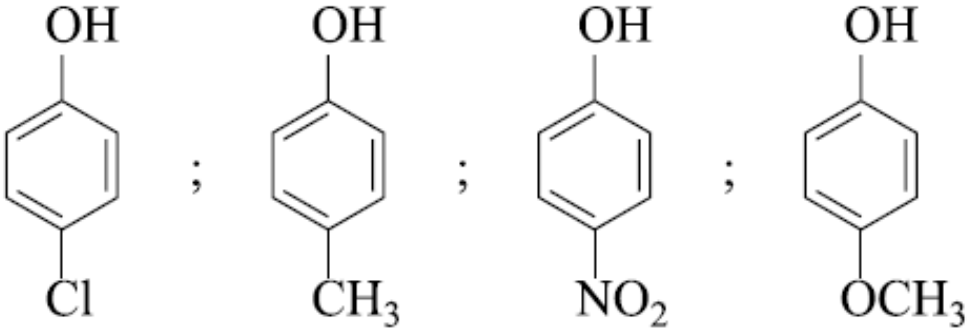
(B) $\mathrm{D} > \mathrm{C} > \mathrm{A} > \mathrm{B}$

(C) $\mathrm{B} > \mathrm{D} > \mathrm{A} > \mathrm{C}$

(D) $\mathrm{A} > \mathrm{B} > \mathrm{D} > \mathrm{C}$

Q.11

Arrange the following compounds in order of decreasing acidity:



(I) (II) (III) (IV)

(A) $\mathrm{II} > \mathrm{IV} > \mathrm{I} > \mathrm{III}$

(B) $\mathrm{I} > \mathrm{II} > \mathrm{III} > \mathrm{IV}$

(C) $\mathrm{III} > \mathrm{I} > \mathrm{II} > \mathrm{IV}$

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(D) IV > III > I > II

Q.12

Which of the following order regarding the acidity of aromatic acids is correct?

(A) benzoic acid > *p*-toluic acid > *p*-hydroxybenzoic acid

(B) benzoic acid > *p*-toluic acid < *p*-hydroxybenzoic acid

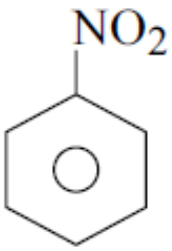
(C) benzoic acid < *p*-toluic acid > *p*-hydroxybenzoic acid

(D) benzoic acid < *p*-toluic acid < *p*-hydroxybenzoic acid

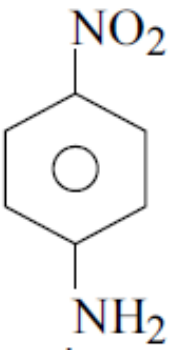
Q.13

Which compound exhibits maximum dipole moment among the following?

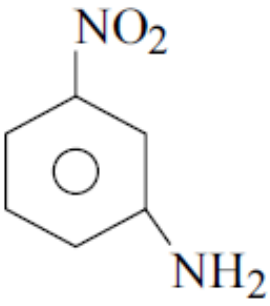
(A)



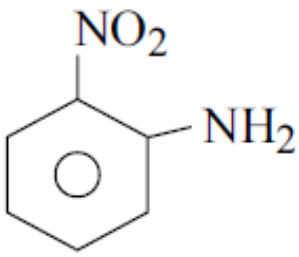
(B)



(C)



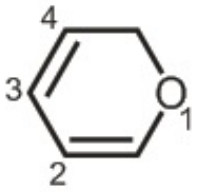
(D)



Q.14

The maximum π-electron-density is present between which of the numbered carbon atoms?

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(A) C1 and C3

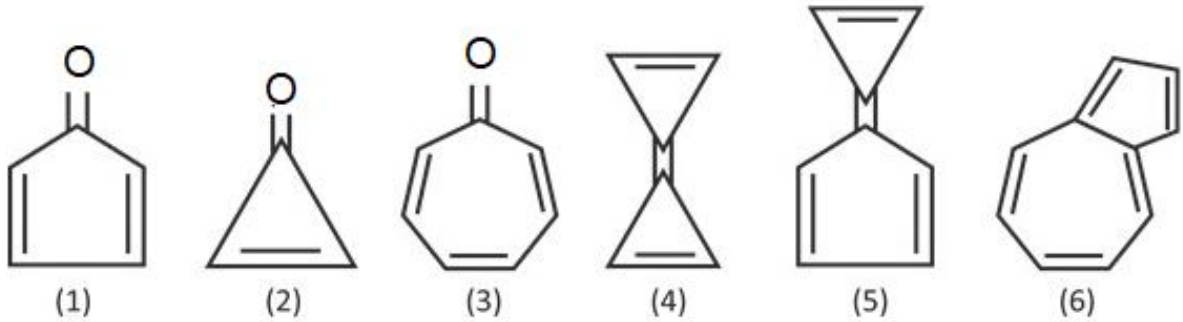
(B) C2 and C4

(C) C2 and C3

(D) C1 and C4

Q.15

Find out total number of compounds which are more stable in its ionic form



(A) 4

(B) 3

(C) 5

(D) 7

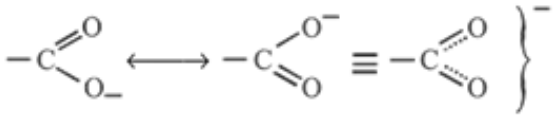
Answers & Solutions

Q.1 Answer:

carboxylate ion gets stabilised by resonance

Solution:

Due to resonance in carboxylate ion, the double bond character of $C=O$ bond in carboxylic acids is greatly reduced as compared to that in aldehydes and ketones.

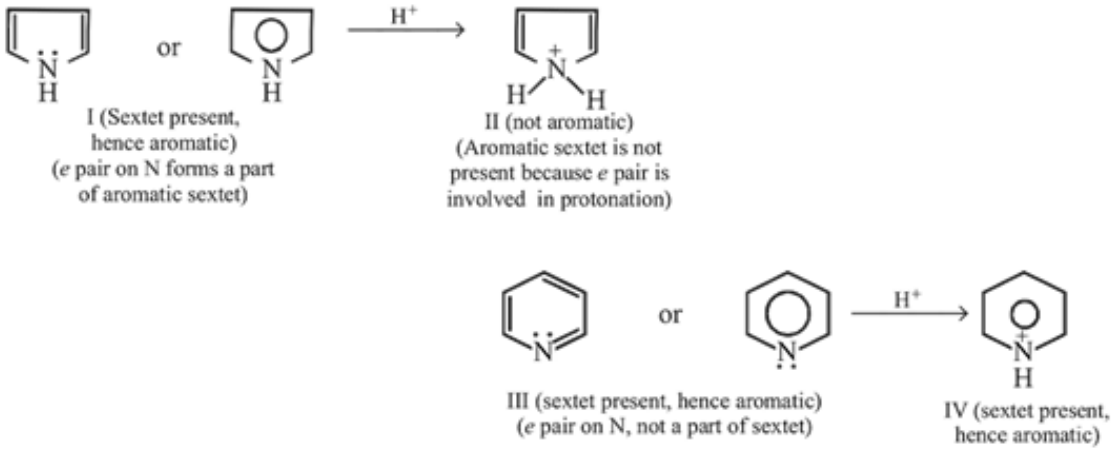


Q.2 Answer:

I, III and IV are aromatic

Solution:

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Q.3 Answer:

(I) > (III) > (II) > (IV)

Solution:

$H_2C = N^+ = N^-$ (I) > $H_2C^- - N^+ \equiv N$ (III) > $H_2C^+ - N = N^-$ (II) > $H_2C^- - N = N^+$ (IV)

I: Octet complete, 6 covalent bond, –ve charge on N

III: Octet complete, 6 covalent bond, –ve charge on C

II: Octet incomplete, 5 covalent bond, –ve charge on N

IV: Octet incomplete, 5 covalent bond, –ve charge on C

Q.4 Answer:

P > Q > R > S

Solution:

The most stable resonating structure is the one with no charge separation. This corresponds to P. This is followed by Q as it has an octet of all atoms complete. None of them is electron deficient. The resonating structure with a positive charge on a relatively electropositive atom and negative charge on the electronegative atom is more stable than the reverse case.

Order of stability is:-

P > Q > R > S

Q.5 Answer:

II is not acceptable because here nitrogen has 10 valence electrons

Solution:

N has no d orbital, hence it can't accommodate more than 8 electrons in its valence shell.

Q.6 Answer:

A

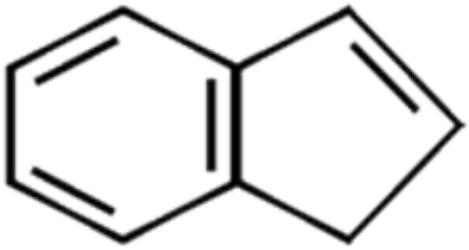
Solution:

The order of stability of resonating structures carrying no charge > carrying minimum charge and

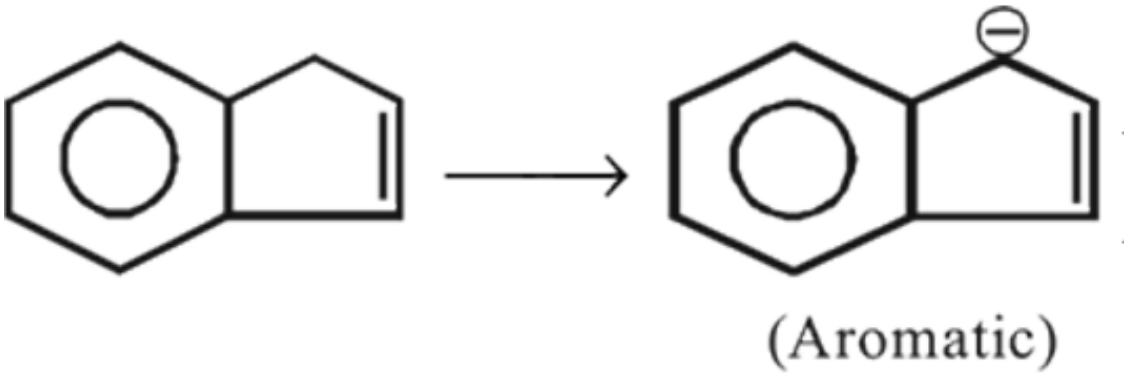
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each atom having octet complete.

Q.7 Answer:



Solution:



Q.8 Answer:

activates ortho and para positions more than the meta position

Solution:

An ortho-para directing group activates all positions of benzene, the ortho and para positions are activated more than meta position.

Q.9 Answer:



Solution:

An aromatic compound has $(4n+2)\pi$-electons, where n is an integer. Cyclopentadiene has only 4π electrons and thus will be nonaromatic.

no conjugation present in cyclopentadiene

Q.10 Answer:

$\mathrm{D} > \mathrm{C} > \mathrm{A} > \mathrm{B}$

Solution:

Electron withdrawing $-\mathrm{NO_2}$ substituent has an acid-strengthening effect while the electron releasing $-\mathrm{CH_3}$ substituent has an acid-weakening effect on phenol. Thus, *p*-cresol is the weakest

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acid amongst the given compounds. The m - and p-nitrophenols are stronger acid than phenol (p-isomer is more strong than m-isomers). The given choice (b) satisfies this requirement.

Q.11 Answer:

III > I > II > IV

Solution:

Electron-releasing group decreases the acidity while electron-attracting group increases the acidity. Thus, the molecule III will have maximum acidity and the molecule IV will have minimum acidity. Thus, the choice (c) is correct.

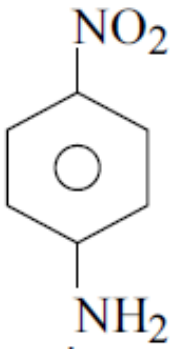
Q.12 Answer:

benzoic acid > p-toluic acid > p-hydroxybenzoic acid

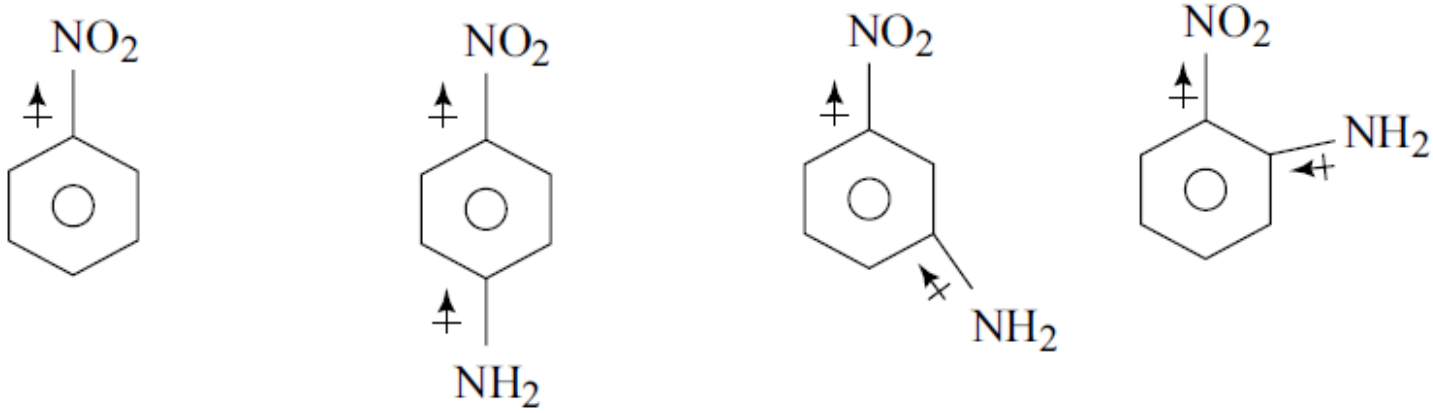
Solution:

Electron releasing group decreases the acidity of benzoic acid. Hydroxyl group is more powerful activator than methyl group.

Q.13 Answer:



Solution:

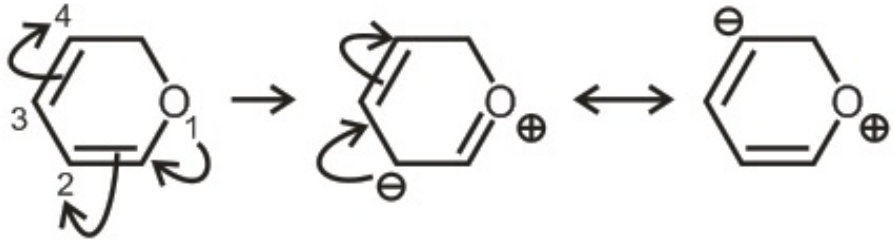


Q.14 Answer:

C2 and C4

Solution:

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π-e^- density is more on C_2 & C_4

Q.15 Answer:

3

Solution:

A molecule is said to be aromatic if the following conditions are satisfied.

An aromatic molecule must be cyclic.

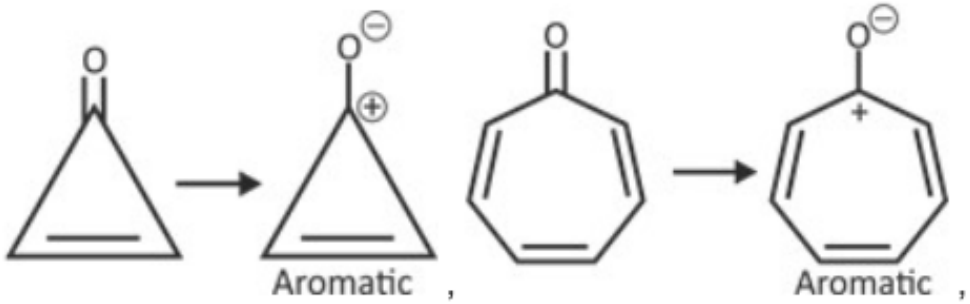
An aromatic molecule must be planar.

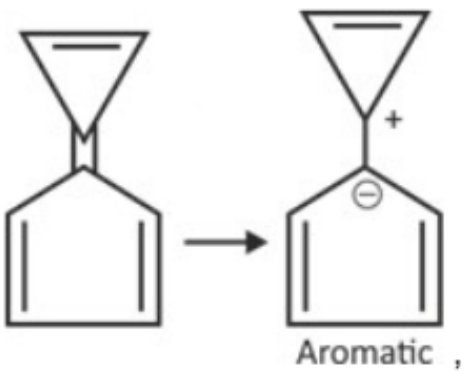
An aromatic ring must contain only sp2-hybridized atoms that can form a delocalised system of π molecular orbitals.

The number of π electrons in the delocalised π system must equal 4n + 2, where n is an integer.

The "4n + 2 rule" was proposed by E. Huckel, and is known as the Hückel rule

Aromatic compounds are more stable compounds among the conjugated compounds.





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